

UNIT-1

Definition of Information Security:

Information Security (InfoSec) refers to the process and practices used to protect data from **unauthorized access, disclosure, modification, destruction, or disruption**. It ensures that the information remains **confidential, accurate, and available** to authorized users.

✔ Importance of Information Security in Today's Digital World:

In the modern digital era, massive amounts of data are stored, processed, and transmitted over networks. Protecting this information is **critical for individuals, businesses, and governments**. Here's why:

1. **Protection from Cyber Attacks:**
 - Prevents hackers from stealing sensitive data such as personal information, passwords, bank details, etc.
2. **Ensures Privacy:**
 - Safeguards the privacy of individuals and organizations by keeping their data confidential.
3. **Maintains Data Integrity:**
 - Ensures that data is not tampered with or altered by unauthorized users.
4. **Ensures Business Continuity:**
 - Helps organizations recover quickly from cyberattacks or data breaches, minimizing downtime and losses.
5. **Compliance with Legal Requirements:**
 - Many laws (e.g., GDPR, HIPAA) require organizations to implement information security measures.
6. **Builds Trust:**
 - Customers and clients are more likely to trust businesses that prioritize data security.

Definition of Network Security:

Network Security is the practice of protecting a computer network and its data from unauthorized access, misuse, modification, destruction, or disruption.

□ Definition:

Network security is a set of technologies, policies, and procedures designed to safeguard the integrity, confidentiality, and availability of computer networks and data.

❑ Objectives of Network Security:

1. **Confidentiality** – Ensuring that sensitive information is accessed only by authorized users.
2. **Integrity** – Ensuring data is accurate and has not been tampered with.
3. **Availability** – Ensuring network services are available when needed.
4. **Authentication** – Verifying the identity of users and devices.
5. **Authorization** – Granting access rights to resources.
6. **Non-repudiation** – Ensuring a sender cannot deny sending a message.

❑ Types of Network Security:

Security Type	Description
Firewall	Filters incoming and outgoing traffic based on rules.
Antivirus & Anti-malware	Detects and removes malicious software.
Intrusion Detection System (IDS)	Monitors traffic to detect suspicious activities.
Intrusion Prevention System (IPS)	Actively blocks detected threats.
Virtual Private Network (VPN)	Secures remote access by encrypting data transmission.
Access Control	Restricts access to network resources.
Data Loss Prevention (DLP)	Prevents unauthorized data sharing.
Network Segmentation	Divides the network into zones to limit access.

❑ Common Threats to Network Security:

- Viruses, Worms, and Trojans
- Ransom ware and Spyware
- Phishing and Social Engineering
- Denial-of-Service (DoS) Attacks
- Man-in-the-Middle (MITM) Attacks
- SQL Injection
- Zero-Day Exploits

Viruses, Worms, and Trojans

- **Virus:** A malicious program that attaches to files and spreads when the file is executed.
 - **Worm:** Self-replicating malware that spreads across networks without needing to attach to a host file.
 - **Trojan Horse:** A deceptive program that appears legitimate but carries hidden malicious functions.
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❑ Ransom ware and Spyware

- **Ransom ware:** Encrypts user data and demands payment to restore access.
- **Spyware:** Secretly gathers user information (like keystrokes, credentials) without consent.

❑ Phishing and Social Engineering

- **Phishing:** Fraudulent emails or messages trick users into revealing personal or financial info.
- **Social Engineering:** Manipulating people into breaking normal security procedures (e.g., pretending to be IT support).

❑ Denial-of-Service (DoS) Attacks

- Overwhelms a server, service, or network with excessive traffic, making it unavailable to legitimate users.

❑ Man-in-the-Middle (MITM) Attacks

- The attacker secretly intercepts and possibly alters communication between two parties without their knowledge (e.g., capturing login credentials during an unencrypted Wi-Fi session).

❑ SQL Injection

- An attacker injects malicious SQL code into input fields (like login forms) to access or manipulate the database.

❑ Zero-Day Exploits

- Attacks that exploit unknown or unpatched vulnerabilities in software before developers can fix them.

Security Attacks:

In today's interconnected world, **information and network security** plays a vital role in protecting sensitive data from unauthorized access, misuse, and destruction.

As digital communication becomes more widespread, the risk of **security attacks** on information systems and networks also increases.

A **security attack** is any attempt to alter, destroy, or gain unauthorized access to data or systems.

Definition of Security Attack

A **security attack** is a deliberate (intentional) act that seeks to compromise the confidentiality, integrity, or availability of information or network resources. These attacks are typically carried out by cybercriminals, hackers, or malicious software (malware).

Classification of Security Attacks

Security attacks are broadly categorized into two main types:

1. Passive Attacks

- Aim to observe or monitor data transmissions.
- Do **not** affect the system resources directly.
- Difficult to detect, as there is no alteration of data.
- **Examples:**
 - Eavesdropping
 - Traffic analysis

2. Active Attacks

- Involve modification, destruction, or interruption of data or services.
- Directly affect system operations and are usually detectable.
- **Examples:**
 - Denial of Service (DoS) attack
 - Man-in-the-Middle (MitM) attack
 - Masquerading (Impersonation)
 - Replay attacks
 - Message modification

Objectives of Security Attacks

- **Data Breach:** Stealing confidential or personal information.
- **Unauthorized Access:** Gaining access to systems or data without permission.
- **System Damage:** Corrupting, deleting, or damaging data and resources.
- **Service Disruption:** Preventing users from accessing services (e.g., DoS).
- **Financial or Reputational Gain:** Attackers often do this to make money, steal secrets, or harm someone's image or business.



Need for Security Awareness

Understanding security attacks is essential to:

- Develop preventive security strategies.
- Implement technologies like encryption, firewalls, and intrusion detection systems.
- Make the security of computer systems stronger and better.

Services & security mechanisms

In Information and Network Security, protecting data and communication is essential. To achieve this, **security services** and **security mechanisms** are used.

1. Security Services

Security services are services that ensure the security of data and communication in a network. These services protect the system against attacks and provide confidentiality, integrity, and availability.

Main Security Services:

1. **Confidentiality**
 - Ensures that information is not accessed by unauthorized users.
 - Achieved through encryption.
2. **Integrity**
 - Ensures data is not altered or tampered with during transmission.
 - Detects any changes made to the data.
3. **Authentication**
 - Verifies the identity of users and systems before allowing access.
 - Example: Login credentials.
4. **Non-repudiation**
 - Prevents denial of actions or messages by the sender or receiver.
 - Ensures accountability.
5. **Access Control**
 - Controls who can access which resources and what actions they can perform.
6. **Availability**
 - Ensures that information and services are available to authorized users when needed.



2. Security Mechanisms

Security mechanisms are tools and techniques used to implement the above services. They help in detecting, preventing, and recovering from security breaches.

Common Security Mechanisms:

1. **Encryption**
 - Converts data into unreadable form to ensure confidentiality.
2. **Digital Signatures**
 - Ensure authenticity and integrity of messages.
3. **Authentication Protocols**
 - Used to verify identities (e.g., passwords, biometrics, OTPs).
4. **Firewalls**
 - Filter incoming and outgoing traffic to block unauthorized access.
5. **Intrusion Detection Systems (IDS)**
 - Monitor networks for suspicious activities.
6. **Access Control Lists (ACLs)**
 - Define permissions for users and systems.
7. **Hash Functions**
 - Used to verify data integrity.

CIA – Confidentiality, Integrity, Availability:

The **CIA triad** is the core model used to guide policies and practices in **Information and Network Security**. It stands for **Confidentiality**, **Integrity**, and **Availability**, and helps ensure that information systems are protected from unauthorized access, alteration, and disruptions.

1. Confidentiality

- **Meaning:** Ensures that data is only accessible to authorized users.
- **Goal:** Protect sensitive information from being viewed or stolen by unauthorized users.
- **Methods Used:**
 - Data encryption
 - User authentication (passwords, biometrics)
 - Access control policies

2. Integrity

- **Meaning:** Ensures that data is accurate and not modified by unauthorized parties.
 - **Goal:** Prevent and detect unauthorized changes to data.
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- **Methods Used:**
 - Hash functions
 - Checksums
 - Digital signatures

3. Availability

- **Meaning:** Ensures that data and resources are accessible to authorized users when needed.
- **Goal:** Prevent service disruption or system failure.
- **Methods Used:**
 - Regular backups
 - Firewalls and anti-DoS tools
 - Alternate systems and emergency response plans.

AAA – Authentication, Authorization, Accounting:

The **AAA framework** is a key concept in Information and Network Security. It helps control and monitors user access to network resources and ensures that only legitimate users can access, perform actions, and be tracked accordingly.

1. Authentication

- **Meaning:** Verifies the identity of a user or device before allowing access.
- **Purpose:** To ensure that the person or system is who they claim to be.
- **Examples/Methods:**
 - Usernames and passwords
 - OTP (One-Time Password)
 - Biometrics
 - Smart cards

2. Authorization

- **Meaning:** Determines what an authenticated user is allowed to do.
- **Purpose:** To grant or deny permission to access specific resources or perform certain actions.
- **Examples:**
 - Access to files, applications, or services
 - Role-based permissions (admin, user, guest)

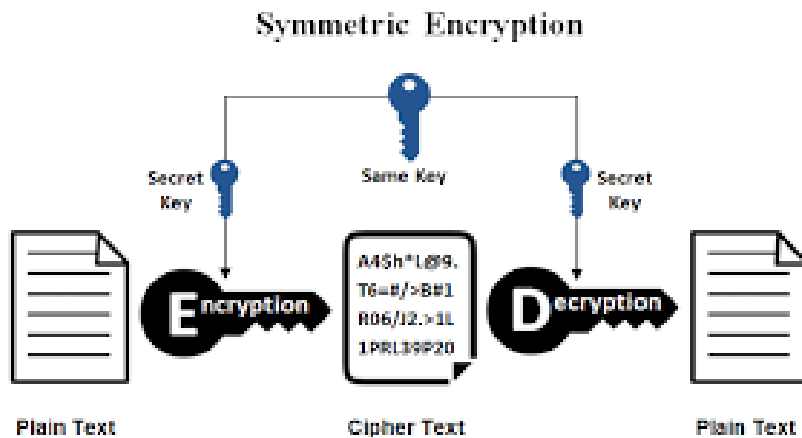


3. Accounting

- **Meaning:** Tracks and records user activities on the network.
- **Purpose:** To keep logs for auditing, billing, or analyzing user behavior.
- **Examples:**
 - Login/logout times
 - Data usage logs
 - Resource access records

Symmetric Cipher Model :

The **Symmetric Cipher Model** is one of the fundamental methods of encryption used to protect data in **Information and Network Security**. In this model, the **same key** is used for both **encryption** and **decryption**.



Features:

- Uses a **single secret key** shared between sender and receiver.
- The **sender** encrypts the plaintext into cipher text using the key.
- The **receiver** decrypts the cipher text back into plaintext using the same key.
- The key must be **kept secret** and securely shared between both parties.

Basic Process:

1. **Plaintext** → (Encryption using secret key) → **Cipher text**
2. **Cipher text** → (Decryption using same key) → **Plaintext**

Advantages:

- Faster and more efficient than asymmetric encryption.
- Suitable for encrypting large amounts of data.

Disadvantages:

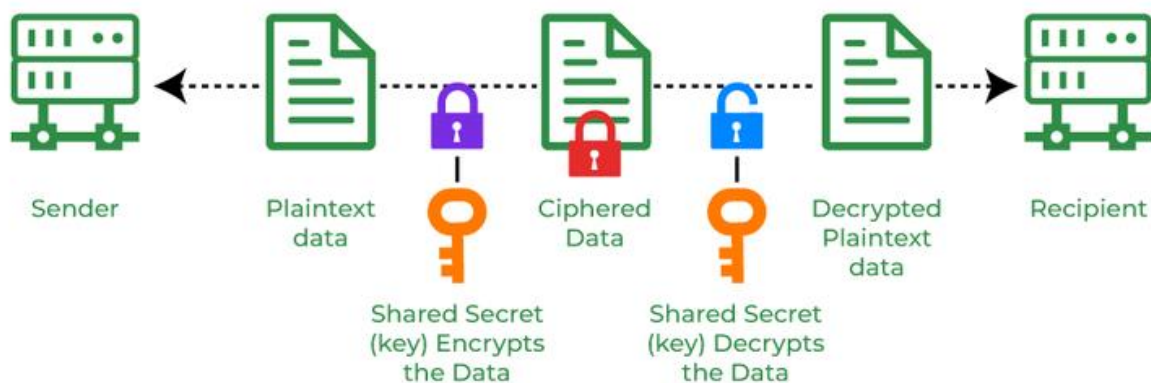
- Key distribution is difficult; both parties must securely share the same key.
- If the key is leaked, the entire system is compromised.

Examples of Symmetric Ciphers:

- DES (Data Encryption Standard)
- AES (Advanced Encryption Standard)
- RC4(Rivest Cipher4), Blowfish(block cipher)

Cryptography :

Cryptography is the science of securing information by converting it into an unreadable format. It plays a vital role in **Information and Network Security** to protect data from unauthorized access and ensure secure communication.



Purpose of Cryptography

- **Confidentiality** – Keeps data secret from unauthorized users.
- **Integrity** – Ensures data is not altered during transmission.
- **Authentication** – Verifies the identity of users or systems.
- **Non-repudiation** – Prevents denial of sending or receiving data.

Basic Terms

- **Plain text:** The original readable message or data.
- **Cipher text:** The encrypted (unreadable) version of the plaintext.
- **Encryption:** Converting plaintext into cipher text using an algorithm.
- **Decryption:** Converting cipher text back into plaintext.

Types of Cryptography

1. **Symmetric Key Cryptography**
 - Same key is used for encryption and decryption.
 - Fast but key sharing is a challenge.
 - Example: AES, DES
2. **Asymmetric Key Cryptography**
 - Uses a pair of keys: **public key** and **private key**.
 - More secure but slower.
 - Example: RSA(Rivest-Shamir-Adleman), ECC(Elliptic Curve Cryptography)

Applications of Cryptography

- Secure email and messaging
- Online banking and e-commerce
- Virtual Private Networks (VPNs)
- Digital signatures and certificates

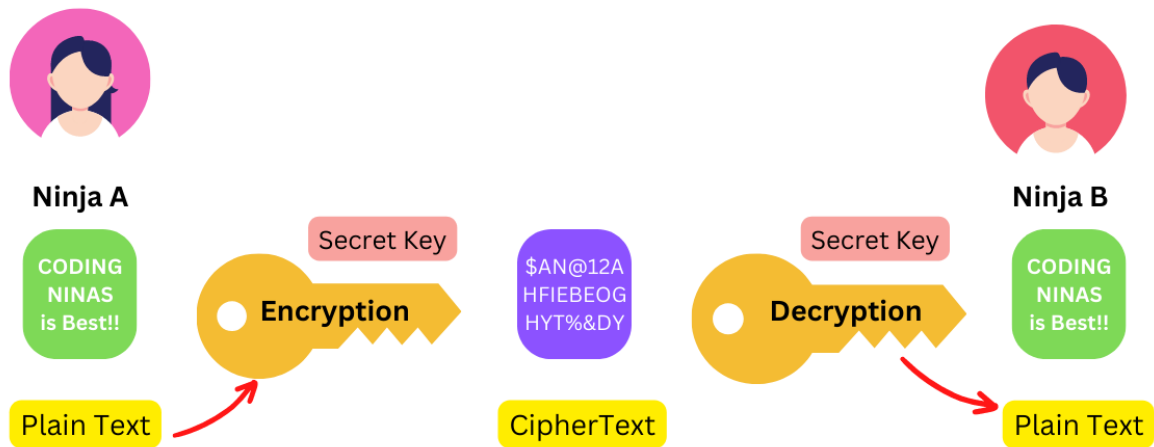
Cryptanalysis and Attacks:

Cryptanalysis

Cryptanalysis is the study and practice of analyzing and breaking cryptographic systems without knowing the secret key. The goal of cryptanalysis is to **decrypt cipher text**, find weaknesses in encryption algorithms, or **recover secret keys**.

It helps in testing the strength of encryption methods and improving security systems.





Types of Cryptanalysis Attacks

1. **Cipher text-Only Attack (COA)**
 - The attacker has only the cipher text.
 - Tries to guess the plaintext or key by analyzing patterns.
2. **Known-Plain text Attack (KPA)**
 - The attacker has both some plaintext and its corresponding cipher text.
 - Uses this information to find the key or decrypt other cipher texts.
3. **Chosen-Plain text Attack (CPA)**
 - The attacker can choose plain texts and get the corresponding cipher texts.
 - Useful in analyzing how the encryption works.
4. **Chosen-Cipher text Attack (CCA)**
 - The attacker chooses cipher texts and obtains their decrypted plaintexts.
 - Aims to find the secret key.
5. **Brute Force Attack**
 - Tries all possible keys until the correct one is found.
 - Time-consuming but always successful if enough time and resources are available.
6. **Side-Channel Attack**
 - Attacks based on information gained from the physical implementation (e.g., timing, power usage, etc.).

Goal of Cryptanalysis

- To break encryption algorithms.
- To test the **security strength** of cryptographic systems.
- To find vulnerabilities and improve **defense mechanisms**.

Compare Cryptography and Cryptanalysis

Aspect	Cryptography	Cryptanalysis
Definition	Cryptography is the science of encrypting and securing information to protect it from unauthorized access.	Cryptanalysis is the science of analyzing and breaking cryptographic systems without knowing the secret key.
Objective	To protect data and ensure secure communication.	To break or weaken the encryption to gain access to the original data.
Focus Area	Designing secure algorithms, keys, and protocols.	Finding vulnerabilities or flaws in encryption methods.
Role	Acts as the defender in a security system.	Acts as the attacker or tester of the system's strength.
Techniques Used	Uses encryption algorithms like AES, RSA, DES, etc.	Uses attack methods like brute force, known-plaintext attack, ciphertext-only attack, etc.
Outcome	Confidential and secure communication.	Exposure of hidden or protected data.
Users	Security professionals, developers, researchers.	Hackers (malicious intent) or ethical hackers (for testing).

Substitution and Transposition Techniques:

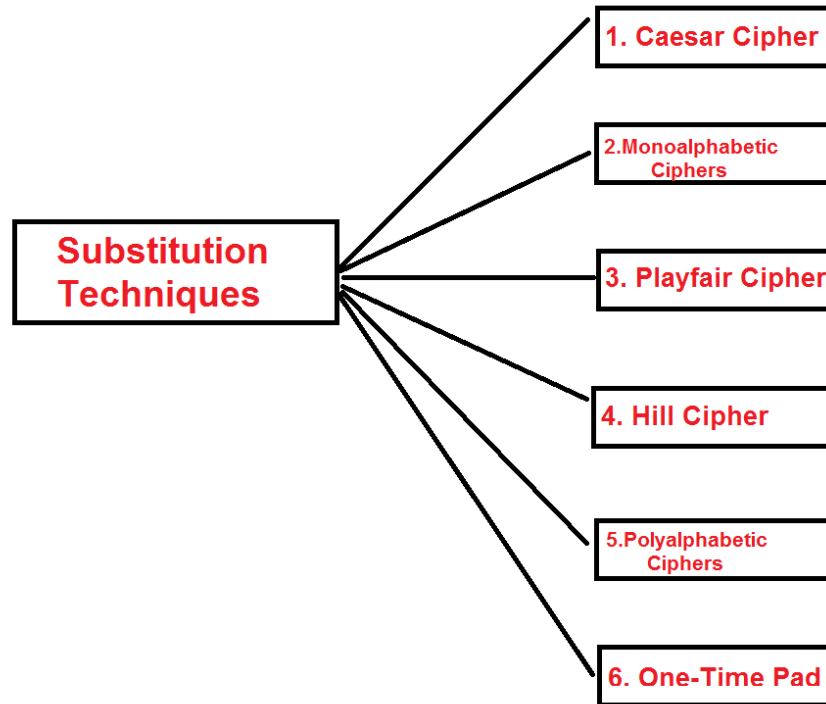
In **cryptography**, the two fundamental techniques used for encryption are **substitution** and **transposition**. These techniques help in securing data by converting readable plaintext into unreadable cipher text.

1. Substitution Technique

- **Definition:** In substitution, each element (character or bit) of the plaintext is **replaced** with another element according to a fixed system or rule.
- The **positions of characters remain unchanged**, only their values are altered.



Types of Substitution Techniques:-



Examples:

- **Caesar Cipher:** Shifts each letter by a fixed number of positions (e.g., $A \rightarrow D$).
- **Mono alphabetic Cipher:** Uses a fixed substitution for each letter.
- **Poly alphabetic Cipher:** Uses multiple substitution alphabets to encrypt the message.

Purpose:

To confuse the relationship between the plaintext and cipher text (called **confusion**).

2. Transposition Technique

- **Definition:** In transposition, the **positions of characters are changed**, but the **characters themselves remain the same**.
- The encryption is done by **reordering** the characters of the plaintext.



Transposition Cipher

enciphering

4	3	1	5	2
N	A	T	S	R
I	S	P	T	O
X	N	I	X	O

plain text: TRANSPOSITION

keyword: 43152

ciphertext: NI

Examples:

- **Rail Fence Cipher:** Writes plaintext in a zigzag pattern and reads it row-wise.
- **Columnar Transposition:** Writes text in rows and reads it in a specified column order.

Purpose:

To spread out the plaintext characters across the cipher text (**diffusion**).



No	Substitution Cipher	Transposition Cipher
1	A substitution technique is one in which the letters of plain text are replaced by other letters or number or symbols. Ex. ABC → XYZ	Transposition cipher does not substitute one symbol for another instead it changes the location of the symbols. EX. ABC → CAB.
2	Monoalphabetic & Polyalphabetic substitution cipher.	Keyless and Keyed transportation cipher.
3	Each letter retains its position changes its identity	Each letter retains its identity but changes its position
4	Ceaser Cipher, Hill cipher, One time pad, Play fair cipher	Rail fence Cipher, Columnar transposition cipher
5	Confusion is commonly carried out through substitution	Diffusion is carried out by using transposition